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           \|||||||||||||||||||/
         --- Welcome to Maude ---
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Maude 3.5.1 built: Jul 16 2025 12:00:00
 Copyright 1997-2025 SRI International
       Tue May 19 12:33:32 2026
```

IDENTITY

reduce in SYSTEMF : *typeCheckF*(*nil*, lambdaT t(1) lambda "x" : t(1) "x", forall t(1) t(1) -> t(1)) .
rewrites: 9 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Bool: *true*

IDENTITY 2

reduce in SYSTEMF : *typeCheckF*(*nil*, app lambdaT t(1) lambda "x" : t(1) "x" t(2) -> t(2), (t(2) -> t(2)) -> t(2) -> t(2)) .
rewrites: 14 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Bool: *true*

DOUBLING FUNCTION

reduce in SYSTEMF : *typeCheckF*(*nil*, lambdaT t(1) lambda "f" : t(1) -> t(1) lambda "x" : t(1) app "f" app "f" "x", forall t(1) (t(1) -> t(1)) -> t(1) -> t(1)) .
rewrites: 31 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Bool: *true*

BOOLEAN RELATED EXAMPLES

reduce in SYSTEMF : *typeCheckF*(*nil*, lambdaT t(1) lambda "x" : t(1) lambda "y" : t(1) "x", forall t(1) t(1) -> t(1) -> t(1)) .
rewrites: 14 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Bool: *true*

reduce in SYSTEMF : *typeCheckF*(*nil*, lambda "b" : forall t(1) t(1) -> t(1) -> t(1) lambdaT t(1) lambda "x" : t(1) lambda "y" : t(1) app app app "b" t(1) "y" "x", (forall t(1) t(1) -> t(1) -> t(1)) -> forall t(1) t(1) -> t(1) -> t(1)) .
rewrites: 47 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Bool: *true*

CHURCH NUMERALS EXAMPLES

reduce in SYSTEMF : *typeCheckF*(*nil*, lambdaT t(1) lambda "s" : t(1) -> t(1) lambda "z" : t(1) "z", forall t(1) (t(1) -> t(1)) -> t(1) -> t(1)) .
rewrites: 13 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Bool: *true*

reduce in SYSTEMF : *typeCheckF*(*nil*, lambdaT t(1) lambda "s" : t(1) -> t(1) lambda "z" : t(1) app "s" "z", forall t(1) (t(1) -> t(1)) -> t(1) -> t(1)) .
rewrites: 22 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Bool: *true*

reduce in SYSTEMF : *typeCheckF*(*nil*, lambdaT t(1) lambda "s" : t(1) -> t(1) lambda "z" : t(1) app "s" app "s" "z", forall t(1) (t(1) -> t(1)) -> t(1) -> t(1)) .
rewrites: 31 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Bool: *true*

TREE EXAMPLES — only used check, but does not really matter

reduce in SYSTEMF **:** *check*(*nil*, lambdaT t(2) lambda "u" : (forall t(1) t(1) -> t(1) -> t(1)) -> t(2) lambda "v" : (forall t(1) t(1) -> t(1) -> t(1)) -> t(2) -> t(2) -> t(2) app "u" lambdaT t(1) lambda "x" : t(1) lambda "y" : t(1) "y") .
rewrites: 29 **in** 0 **ms cpu (**0 **ms real) (~ rewrites/second)**
result Type: forall t(2) ((forall t(1) t(1) -> t(1) -> t(1)) -> t(2)) -> ((forall t(1) t(1) -> t(1) -> t(1)) -> t(2) -> t(2) -> t(2)) -> t(2)

reduce in SYSTEMF **:** *check*(*nil*, lambdaT t(2) lambda "u" : (forall t(1) t(1) -> t(1) -> t(1)) -> t(2) lambda "v" : (forall t(1) t(1) -> t(1) -> t(1)) -> t(2) -> t(2) -> t(2) app app app "v" lambdaT t(1) lambda "x" : t(1) lambda "y" : t(1) "x" app "u" lambdaT t(1) lambda "x" : t(1) lambda "y" : t(1) "y" app "u" lambdaT t(1) lambda "x" : t(1) lambda "y" : t(1) "x") .
rewrites: 82 **in** 0 **ms cpu (**0 **ms real) (~ rewrites/second)**
result Type: forall t(2) ((forall t(1) t(1) -> t(1) -> t(1)) -> t(2)) -> ((forall t(1) t(1) -> t(1) -> t(1)) -> t(2) -> t(2) -> t(2)) -> t(2)

reduce in SYSTEMF **:** *check*(*nil*, lambdaT t(2) lambda "u" : (forall t(1) t(1) -> t(1) -> t(1)) -> t(2) lambda "v" : (forall t(1) t(1) -> t(1) -> t(1)) -> t(2) -> t(2) -> t(2) app app app "v" lambdaT t(1) lambda "x" : t(1) lambda "y" : t(1) "x" app "u" lambdaT t(1) lambda "x" : t(1) lambda "y" : t(1) "x" app app app "v" lambdaT t(1) lambda "x" : t(1) lambda "y" : t(1) "y" app "u" lambdaT t(1) lambda "x" : t(1) lambda "y" : t(1) "x" app "u" lambdaT t(1) lambda "x" : t(1) lambda "y" : t(1) "y") .
rewrites: 134 **in** 0 **ms cpu (**0 **ms real) (~ rewrites/second)**
result Type: forall t(2) ((forall t(1) t(1) -> t(1) -> t(1)) -> t(2)) -> ((forall t(1) t(1) -> t(1) -> t(1)) -> t(2) -> t(2) -> t(2)) -> t(2)

Bye.